popquiz telescope intro

This is a preview of the published version of the quiz

Started: Mar 22 at 7:26pm

Quiz Instructions

https://phet.colorado.edu/sims/html/geometric-optics/latest/geometric-optics_all.html
(https://phet.colorado.edu/sims/html/geometric-optics/latest/geometric-optics_all.html)

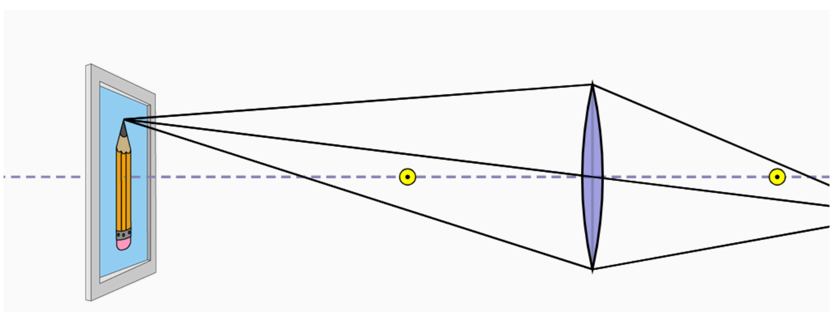
Open this app

Question 1 20 pts

Go to:

https://phet.colorado.edu/sims/html/geometric-optics/latest/geometric-optics_all.html
(https://phet.colorado.edu/sims/html/geometric-optics/latest/geometric-optics_all.html)

Select a converging lens (like those used for magnifier glasses). You will learn the principles of a refractive telescope.



The yellow dots are called the focal points of the lens. On the right you have the image. You are placing the lens in front of the image (like a tree far away or an EXIT sign) and you get an image on a screen (on in your eye).

Move the object back and forth . **!!! The object stays to the left of the focal point !!! (not between the dot and the lens).**

What is true

- [ ] The image is upside dow
- [ ] The image is upright

○ It depends on the position of the object

○ There is no image.

Question 2 20 pts

As you move the object farther and farther away from the lens. (like a star would be)

○ the image gets smaller

○ the image gets larger

○ it gets smaller and larger

○ it gets larger and smaller

Question 3 20 pts

In the upper corner. Click on the star. As you move the star away from the lens. To heaven.

○ the image move closer to the focal point on the other side of the lens

○ as the object (left) moves to the left. The image moves to the right

○ There is no image

○ The image fades away

Question 4 20 pts

PLace the star far away to the left (heaven).

select many rays in the Lower left. The lens is like a bucket. It collect photons. more rays you see more photons collected. Change the diameter of the lens. increase, decrease.

What is true

- [ ] as the diameter increases, the image gets brighter. more rays
- [ ] as the diameter increases, the image gets brighter. less rays
- [ ] the brightness is not size dependent
- [ ] smaller diameter, brighter is the image.

Question 5 20 pts

Down the page click on mirror. Its a concave mirror that behaves like a converging lens.

As you move the object farther and farther away from the mirror. The image

- [ ] gets smaller and closer to the focal point
- [ ] gets larger and closer to the focal point
- [ ] gets smaller and is upright
- [ ] there is no image

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